Example 3.66 If $A=\begin{pmatrix}1&8&3\\3&5&0\\8&7&6\end{pmatrix}$, $B=\begin{pmatrix}8&-6&-4\\2&11&-3\\0&1&5\end{pmatrix}$, $C=\begin{pmatrix}5&3&0\\-1&-7&2\\1&4&3\end{pmatrix}$

compute the following : (i) $3A+2B-C$ (ii) $\frac{1}{2}A-\frac{3}{2}B$

Solution (i) $3A+2B-C=3\begin{pmatrix}1&8&3\\3&5&0\\8&7&6\end{pmatrix}+2\begin{pmatrix}8&-6&-4\\2&11&-3\\0&1&5\end{pmatrix}-\begin{pmatrix}5&3&0\\-1&-7&2\\1&4&3\end{pmatrix}$

$=\begin{pmatrix}3&24&9\\9&15&0\\24&21&18\end{pmatrix}+\begin{pmatrix}16&-12&-8\\4&22&-6\\0&2&10\end{pmatrix}+\begin{pmatrix}-5&-3&0\\1&7&-2\\-1&-4&-3\end{pmatrix}$

$=\begin{pmatrix}14&9&1\\14&44&-8\\23&19&25\end{pmatrix}$

(ii) $\frac{1}{2}A-\frac{3}{2}B=\frac{1}{2}(A-3B)$

$=\frac{1}{2}\left(\begin{pmatrix}1&8&3\\3&5&0\\8&7&6\end{pmatrix}-3\begin{pmatrix}8&-6&-4\\2&11&-3\\0&1&5\end{pmatrix}\right)$

$=\frac{1}{2}\left(\begin{pmatrix}1&8&3\\3&5&0\\8&7&6\end{pmatrix}+\begin{pmatrix}-24&18&12\\-6&-33&9\\0&-3&-15\end{pmatrix}\right)=\frac{1}{2}\begin{pmatrix}-23&26&15\\-3&-28&9\\8&4&-9\end{pmatrix}$

$=\begin{pmatrix}-\frac{23}{2}&13&\frac{15}{2}\\-\frac{3}{2}&-14&\frac{9}{2}\\4&2&-\frac{9}{2}\end{pmatrix}$

Exercise 3.18

1. If $A=\begin{pmatrix}1&9\\3&4\\8&-3\end{pmatrix}$, $B=\begin{pmatrix}5&7\\3&3\\1&0\end{pmatrix}$ then verify that

   (i) $A+B=B+A$ (ii) $A+(-A)=(-A)+A=O$.

2. If $A=\begin{pmatrix}4&3&1\\2&3&-8\\1&0&-4\end{pmatrix}$, $B=\begin{pmatrix}2&3&4\\1&9&2\\-7&1&-1\end{pmatrix}$ and $C=\begin{pmatrix}8&3&4\\1&-2&3\\2&4&-1\end{pmatrix}$ then verify that

   $A+(B+C)=(A+B)+C$.

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